# THE CHINESE ECONOMY

## A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 3: How do economists estimate causal effects? From Mao to the reforms

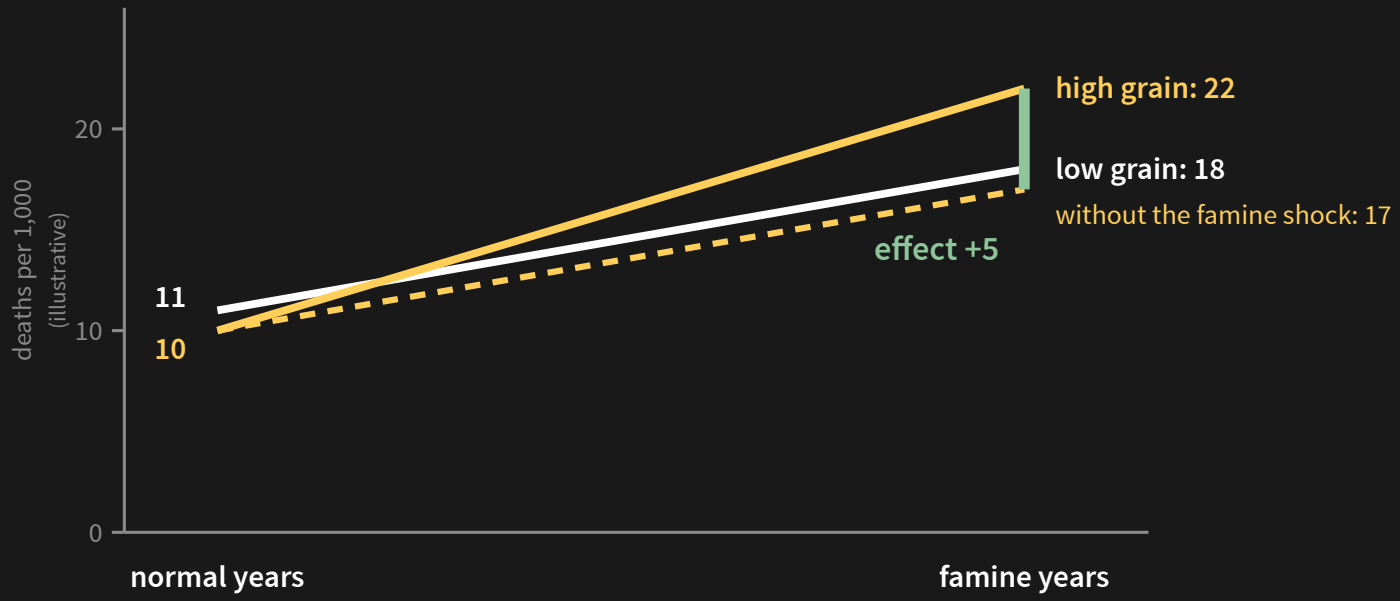
# LAST WEEK, BY HAND

On the board: the table of last week. Deaths per 1,000 people, illustrative numbers.

| | Normal years | Famine years | Change |
|---|---|---|---|
| **High** expected grain | 10 | 22 | +12 |
| **Low** expected grain | 11 | 18 | +7 |

**Difference in differences: +12 − 7 = +5. But why may we subtract the low-grain change?**

# THE ASSUMPTION




*Parallel trends*: without the treatment, the two groups would have ***moved together***.
Their gap may exist, but — net of the treatment — it would stay **fixed over time**.

It is an **Assumption**: it has to be true if we want to retrieve a causal effect.

# 1961-76 — READJUSTMENT, THIRD FRONT, CULTURAL REVOLUTION

- **1961-63** readjustment: investment chopped back, about 20 million workers sent to the countryside
- **1964-66** Third Front: a new expansion into the interior
- **1966-76** Cultural Revolution: 17 million urban school-leavers sent to the countryside (*xiafang*); schools and universities closed
- **1976**: Mao dies

# 1968-78: YOUNG PEOPLE SENT TO THE COUNTRYSIDE

- About **16-17 million urban youths** were sent to live and work in villages during the Cultural Revolution
- Who went was not random: **family background**, schooling, politics, and household rules (often one child could stay in the city)
- Most came back after 1978

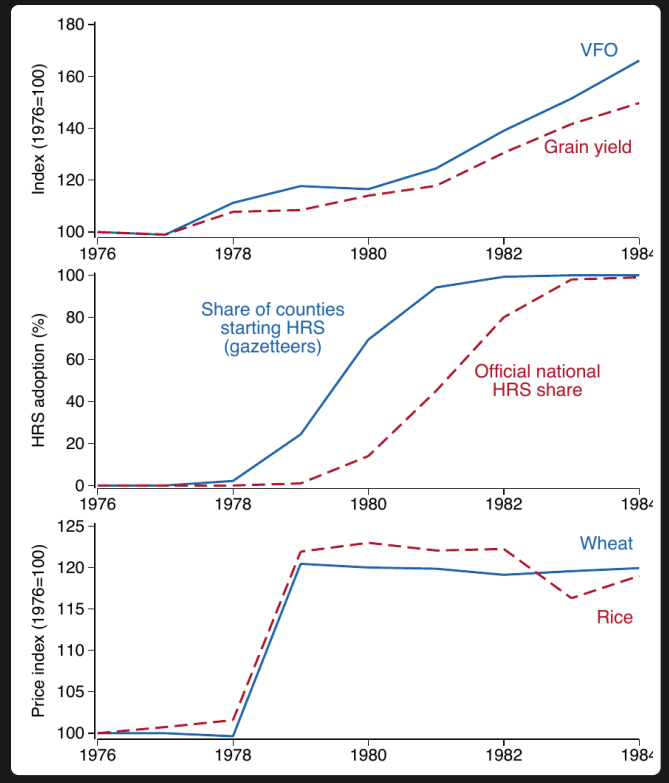
## Question: did the years in the countryside change their lives?

# 1976-78 — THE TURN

- Growth had accelerated under the Big Push — but even more in Taiwan and Hong Kong, far ahead of the PRC by 1978
- That demonstration effect convinced Chinese leaders to reform and open up
- **December 1978**: Third Plenum — ease the terms of trade with agriculture, "give farmers a chance to catch their breath"
- Deng Xiaoping consolidates power; *gaige kaifang*, reform and opening

# 1978-84: THE LAND GOES BACK TO HOUSEHOLDS

- **Household Responsibility System**: the collective's land is contracted to families; above the quota, they keep what they grow
- First experiments: **Anhui, 1978**, after a severe drought. Then province by province, county by county, until 1984
- Meanwhile, **1979**: the state raises the price it pays for quota grain by about 20%
- Official grain yields: **+43%** from 1978 to 1984
- The reform did not arrive everywhere at once: a **staggered rollout**. For an economist, a gift




Ferguson and Kim (2026), Fig. 1: farm output and yields; share of counties that started the reform; procurement prices (1976 = 100)

# WHAT WAS THE CAUSAL EFFECT OF THE TRANSITION?

- The ideal experiment: **two Chinas** in 1978, identical; one reforms, one does not
- There is **one** China. And everything changed **at once**: land, prices, leaders, trade, the opening to the world
- No comparison group, no lottery, no counterfactual we can approach

*Whereof one cannot speak, thereof one must be silent.*

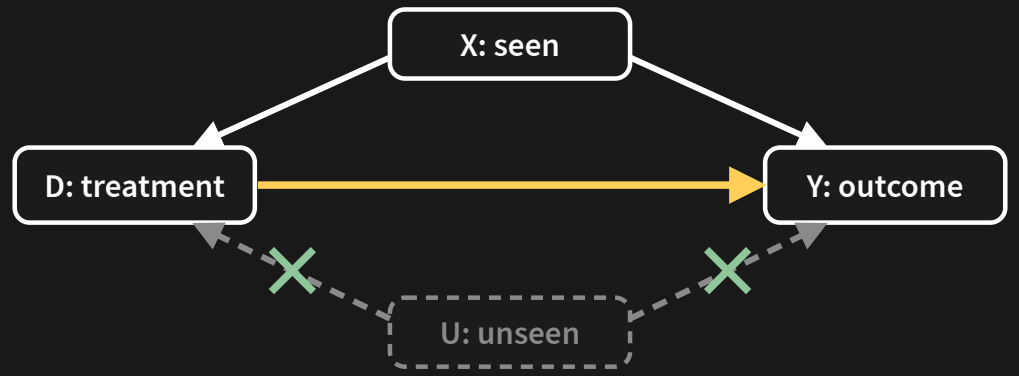
Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, proposition 7 (1921; English translation by C. K. Ogden, 1922)

**…unless we cut the question into pieces we can answer**

# PIECES WE CAN ANSWER

| Question | Who could be compared with whom? | Method | Paper |
|---|---|---|---|
| Did the years in the countryside change the sent-down youths' lives? | brothers and sisters: one sent, one not | **matching** | Xie, Jiang, Greenman, "Did Send-down Experience Benefit Youth?" (2008) |
| Did giving the land back to households change who was born? | families in the same county and year, before and after the reform | **difference in differences**, year by year; nature's lottery | Almond, Li, Zhang, "Land Reform and Sex Selection in China" (2019) |
| Did it raise yields? | early and late reformers alike in what we see; fields just across a provincial border, before and after | **regression, propensity score**; difference in differences at the border | Ferguson, Kim, "What Started China's Miracle? The View from Outer Space" (2026) |

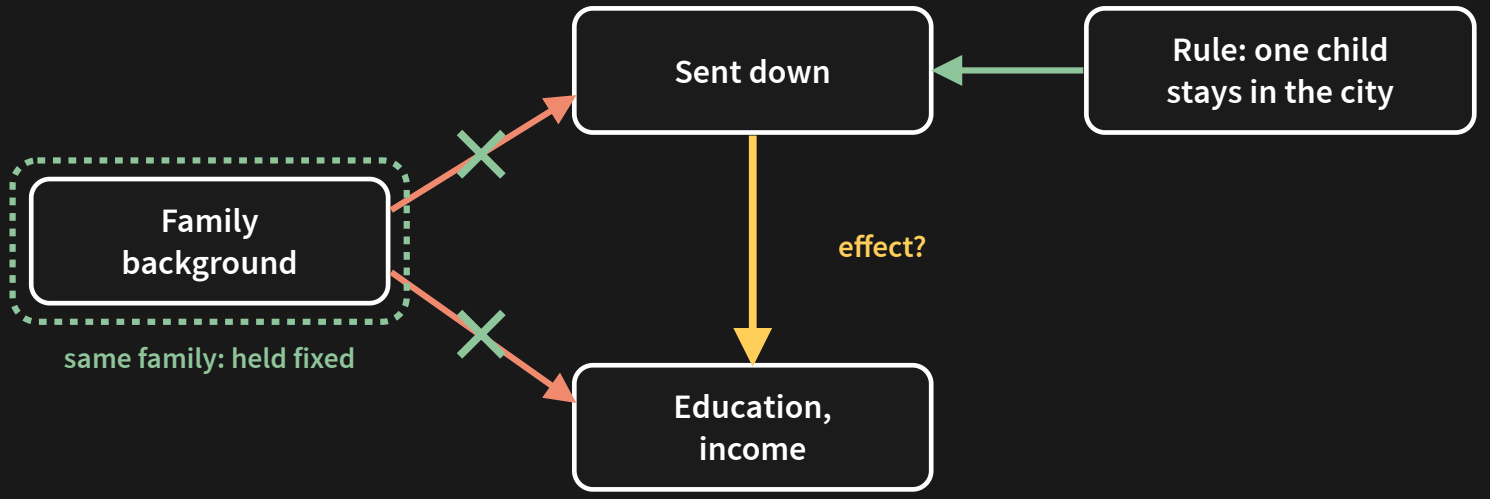
# THREE WAYS TO CLOSE THE BACK DOOR




DiD: U cancels, if it does not change over time

| Strategy | What it does in the graph | What we must assume |
|---|---|---|
| **Lottery** | cuts every arrow into D | the draw is really random |
| **Matching** | holds fixed the X we can see: compare like with like | no back door runs through what we cannot see |
| **Difference in differences** | lets an unseen U through, if it does not change over time | parallel trends |

# MATCHING: COMPARE LIKE WITH LIKE



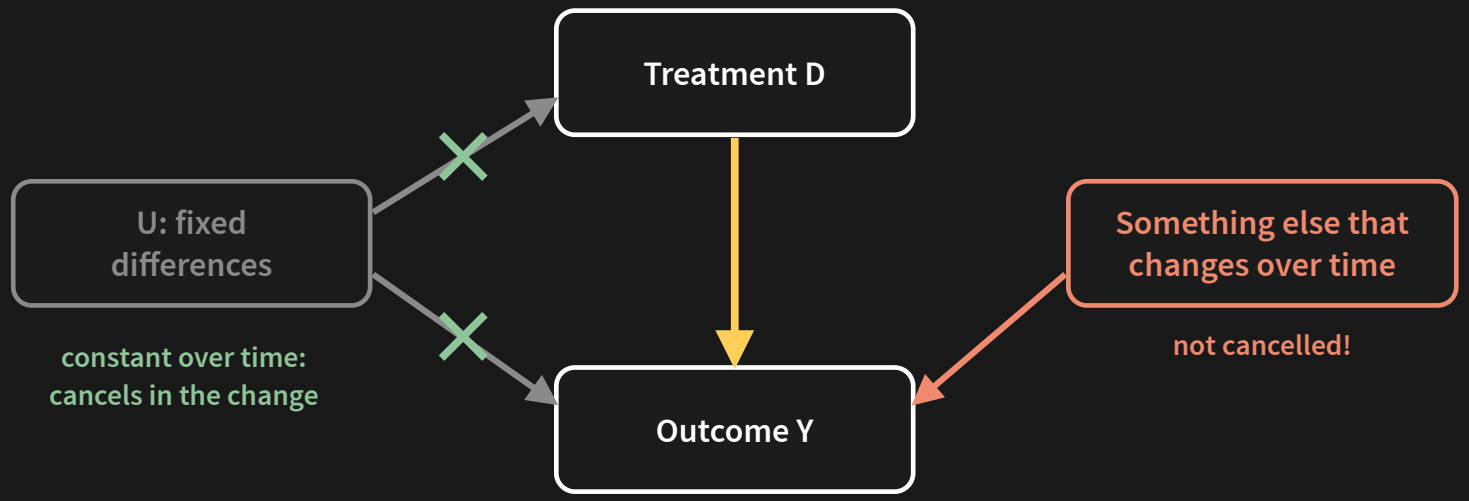

Compare a sent-down youth with someone **as similar as possible** who stayed: same city, same age... ideally **the same family**

# SENT-DOWN YOUTHS: TWO COMPARISONS

- **Naive**: among people who were sent down for a short time, **15.2%** reached college; among those never sent down, **10.9%**. Was the countryside good for them?
- **Siblings**: 344 pairs of brothers and sisters, one sent down and one not. **No discernible difference** in any of four measures of socioeconomic status
- The first gap was **selection**, not an effect

Xie, Jiang and Greenman (2008), *Social Science Research* 37(2): survey of 1999 in Shanghai, Wuhan and Xi'an

# DIFFERENCE IN DIFFERENCES, IN THE GRAPH




DiD survives what we cannot see, **if it stays put**. It fails with what **moves at the same time** as the treatment

# LAND REFORM AND SEX SELECTION IN CHINA

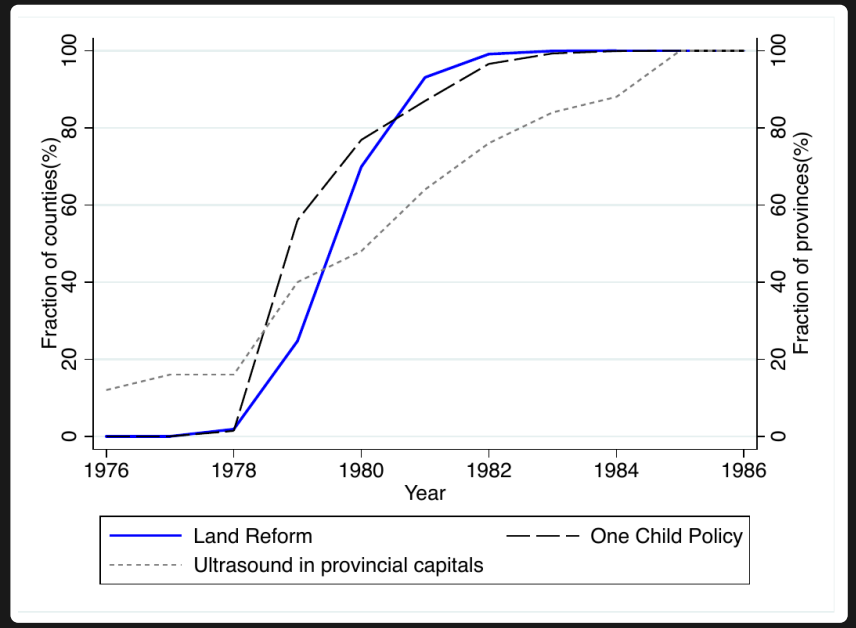
Douglas Almond, Hongbin Li, Shuang Zhang

JPE, 2019

- In a normal population about **105 boys** are born for every 100 girls
- In China the sex ratio at birth rose from **1.06 in 1978** to **1.20 in 2000**: hundreds of thousands of "missing girls" every year
- The usual suspects: ultrasound made sex selection cheaper; families had fewer children (the One Child Policy)
- But the rise **started around 1980**, when ultrasound was rare in the countryside and fertility was roughly flat. **Something else** started then
- The candidate: the **land reform** of 1978-84, which raised rural incomes

# QUESTION AND DATA

- **Question**: did giving land back to households increase sex selection?
- **Treatment**: the year the reform started in each county, from the **county histories** (gazetteers) of 914 counties, half of rural China. The same histories give the year the One Child Policy started
- **Outcome**: the sex of children born 1974-86, from the **1990 census**. Within each family, children are ordered by birth date: first child, second child...
- Few people had moved (hukou): children are counted where they were born




Almond, Li and Zhang, Fig. 2: share of counties that had started land reform (solid) and the One Child Policy (dashed); provinces with ultrasound in the capital (dotted)

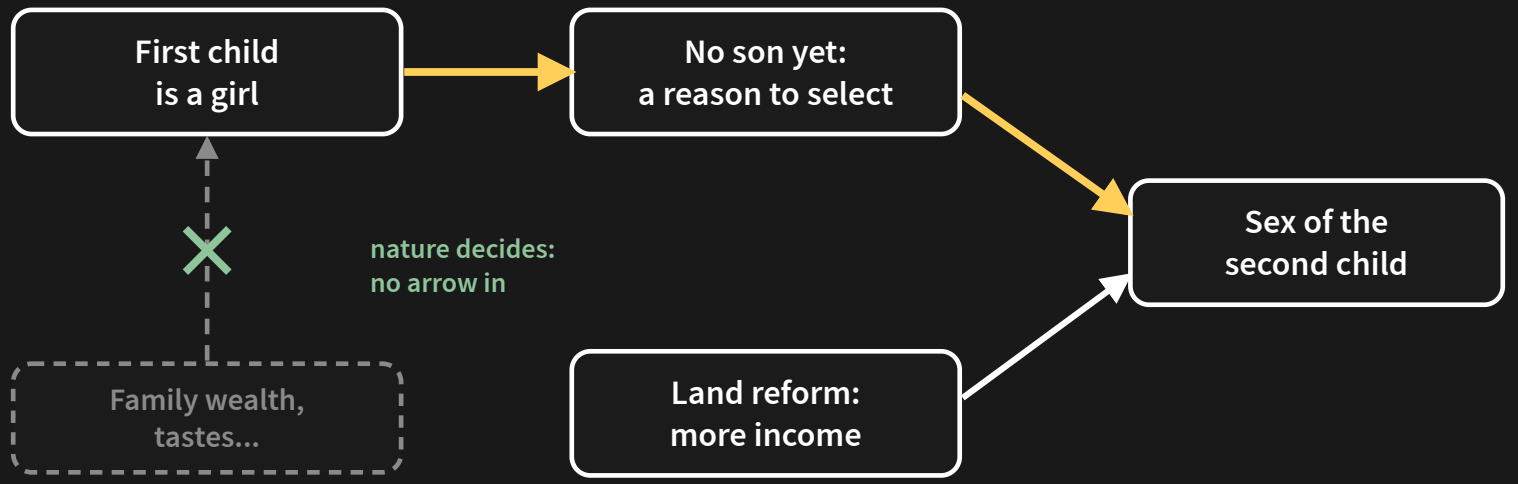
# STRATEGY

Line up each county on **its own** reform year (year 0). Then compare, in the **same county and the same year**, three kinds of births:

| Share of boys among... | Before reform | After reform | Change |
|---|---|---|---|
| first children (nature decides) | 0.51 | 0.51 | 0 |
| second children **after a boy** (little reason to select) | 0.50 | 0.50 | 0 |
| second children **after a girl** (a reason to select) | 0.53 | 0.56 | **+0.03** |

- **Difference in differences**: the change after a girl (+3 points) minus the change after a boy (0). A shock that hit the whole county that year moves both rows
- The assumption: without the reform, girl-first and boy-first families would have kept the **same gap** (parallel trends)
- Year by year instead of before/after: an **event study**. Flat lines before year 0 make the assumption believable

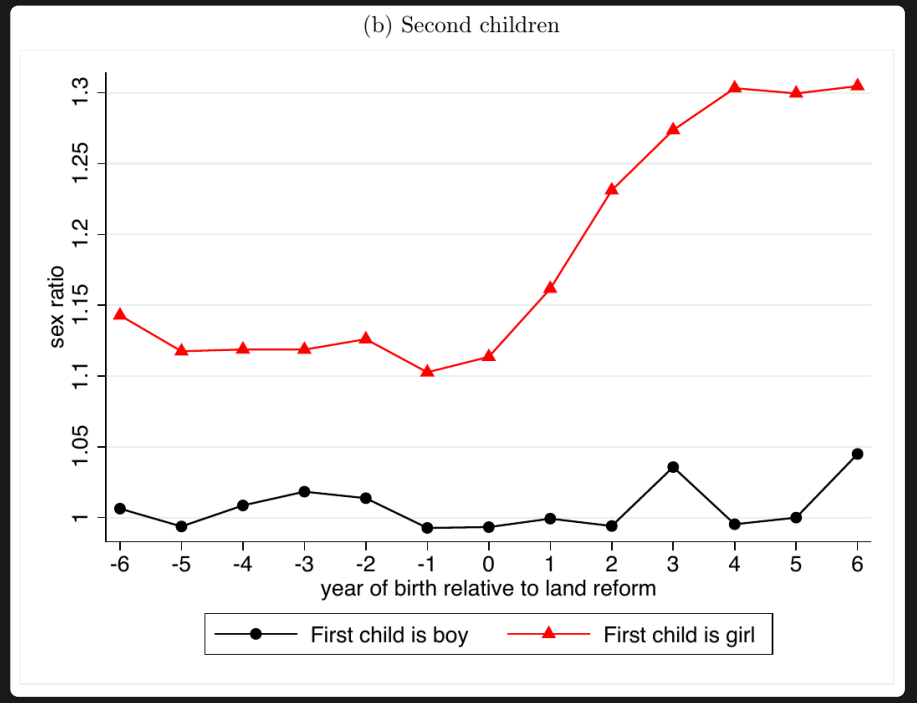
# NATURE'S LOTTERY




- The sex of the first child is decided by **nature**: about 1.05 boys per girl, before and after the reform
- Nothing about the family pushes it one way or the other: no arrow comes in
- It moves the **reason to select**, and touches the second child only through it. That is the logic of an **instrument**
- The threat: families with a first girl being different in other ways. Normal first births (1.05) are the evidence against it

# RESULTS

- Red: second children after a **first girl**. Flat before the reform, then from about **1.1 to 1.3 boys per girl** within four years
- Black: second children after a **first boy**. Nothing happens
- First children (not shown): stable at about 1.05
- Size: about **half** of the rise in rural sex ratios 1978-86, roughly **one million missing girls**




Almond, Li and Zhang, Fig. 4b: sex ratio of second children by year of birth relative to land reform (0 = reform year)

You have already seen an event study: Meng, Qian and Yared's Figure 3, around the famine

# IS IT REALLY THE LAND REFORM?

- **The rival: the One Child Policy**, started at almost the same time. But not in the same year in each county: adding the county's One Child Policy date **does not change** the result, and the rise starts **before** the rural policy
- **Who has second children** did not change with the reform: the result is not a different mix of families
- **Why?** More consistent with **income**: larger effects among more educated parents and where output grew more
- **How?** Ultrasound existed in provincial capitals: larger rise in rural counties **connected by rail** to one; a rise also where none was reachable

# ONE PAPER, THREE TOOLS

| Tool | In Almond, Li and Zhang | What it keeps out |
|---|---|---|
| **Compare like with like** (matching) | births in the same county and the same year | anything that hit a county in a year: weather, a clinic, a campaign |
| **Difference in differences**, year by year | before and after each county's own reform | fixed differences between counties and families |
| **Nature's lottery** (instrument logic) | the sex of the first child | differences between families that want to select and families that do not |

Each problem can be taken apart **separately**, inside the same paper

# WHAT STARTED CHINA'S MIRACLE? THE VIEW FROM OUTER SPACE

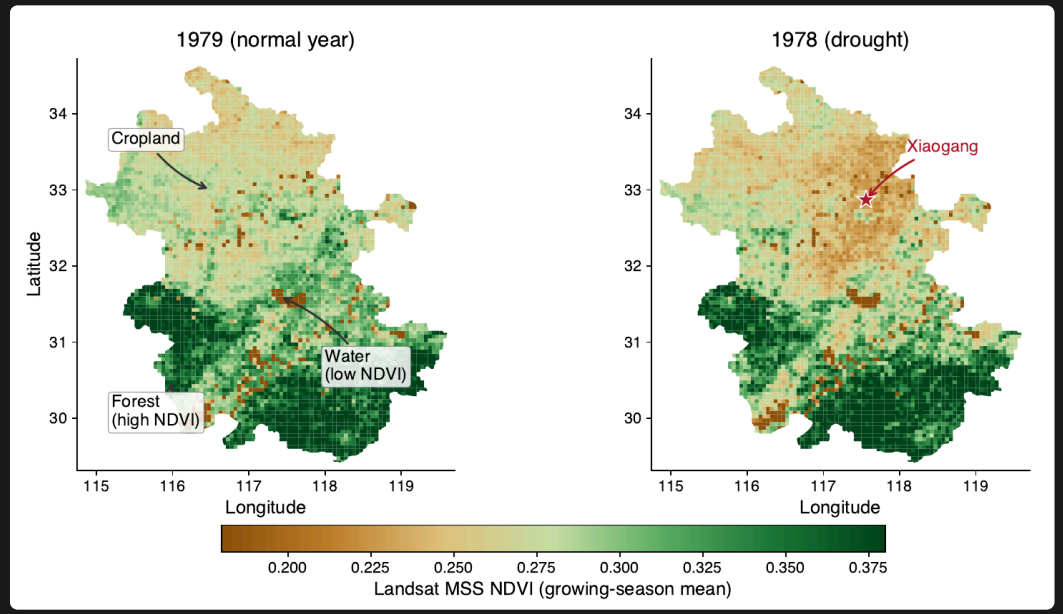
Joel Ferguson, Oliver Kim

Working paper, 2026

- **The standard story**: giving land back to households raised farm productivity and started the miracle. Lin (1992): almost **half** of farm output growth 1978-84
- It rests on **official statistics**, from a state whose local officials had over-reported harvests before, during the Great Leap
- And two problems we can now name: the early reformers were **poor and drought-hit**; prices **rose in 1979** for everybody
- **Question**: did reforming **sooner** raise grain **yields**?

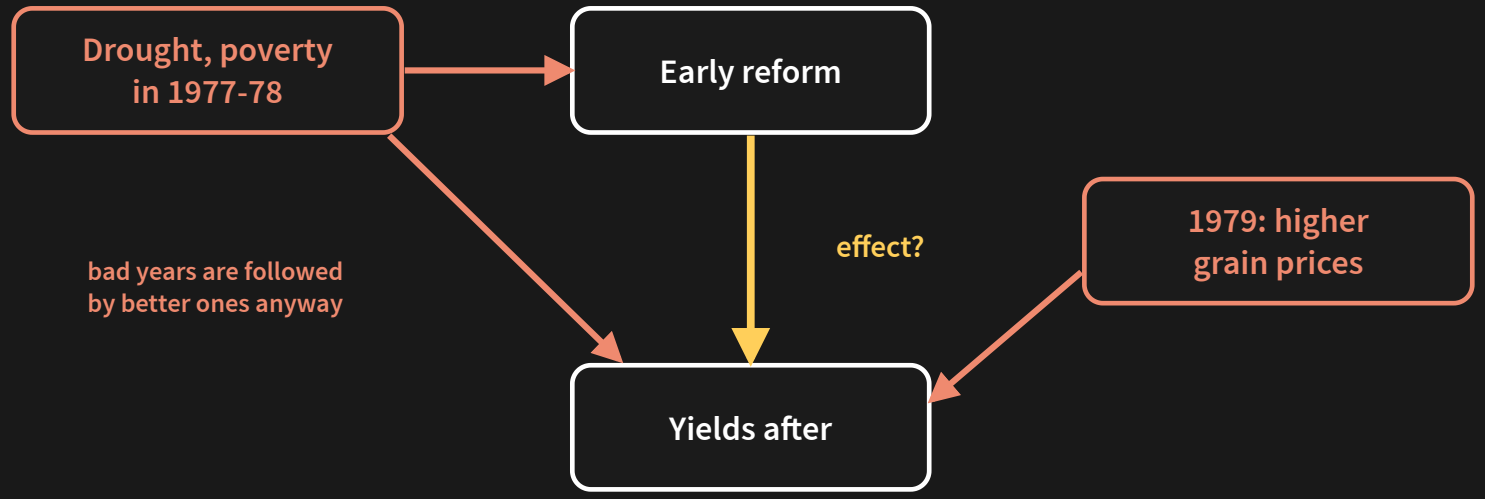
# DATA FROM SPACE

- **Satellites**: Landsat (from 1972, dots of about 60 metres) and a coarser weather satellite
- **NDVI**: how green the fields are. Checked against crop yields in many countries and periods
- **Independent** of the Chinese statistical office
- Reform dates: official, by province (Lin 1992), and from county histories (Almond, Li and Zhang)




Ferguson and Kim, Fig. 2: Anhui seen by Landsat in 1979 (normal year) and 1978 (drought). The star: Xiaogang, where households started the reform

# DID THE REFORM RAISE YIELDS? DRAW THE GRAPH FIRST

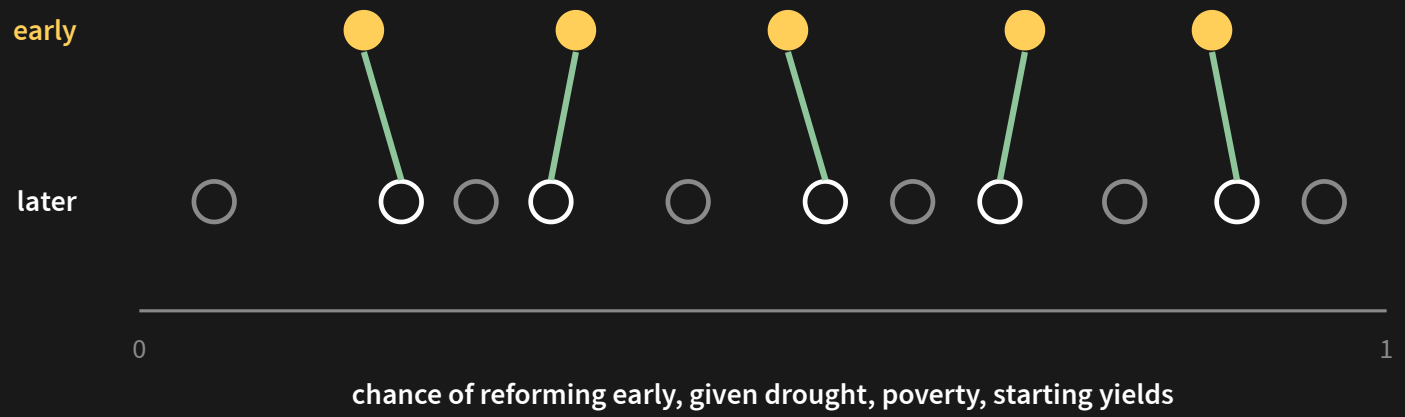



- The first to reform were the **poorest and drought-hit**: their yields would have bounced back anyway (**mean reversion**)
- And in **1979 prices rose for everybody**, at the same time as the reform spread

# HOLDING MANY THINGS FIXED: REGRESSION, FOR DUMMIES

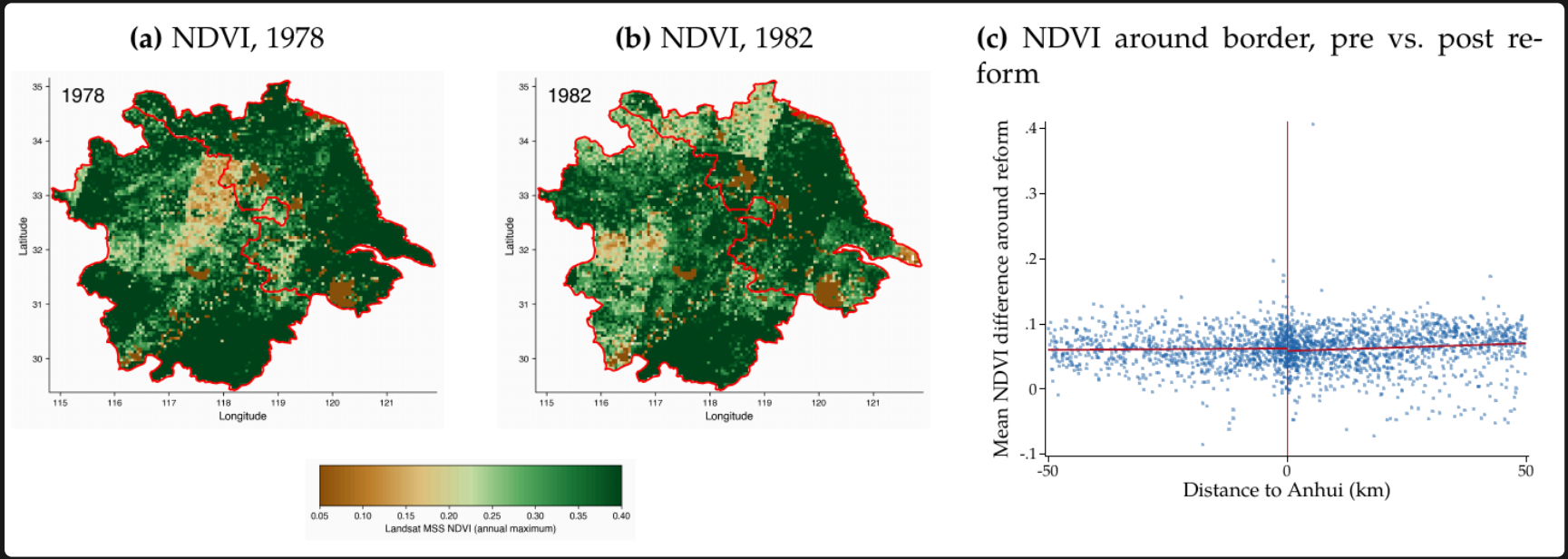
- Early and late reformers differ in **many** things we can measure: starting yields, drought, poverty, crops...
- Ideal: compare early and late reformers that are **identical** in all of them. With a handful of things, the groups empty fast (remember: 19 provinces, one per cell)
- **Regression** fills the empty cells by assuming a simple shape: each factor shifts yields by a fixed amount. It compares like with like, **on average**
- It only holds fixed **what we measure**. The luck of a good year, local politics, the 1979 prices: only if measured

# PROPENSITY SCORE, FOR DUMMIES




- **Step 1**: from what we see, compute each county's **chance** of reforming early. Many things become **one number**
- **Step 2**: pair each early reformer with a later one that had a **similar chance**, and compare their yields
- Same assumption as matching: **nothing unseen** drives both timing and yields. Ferguson and Kim go further: **borders** and **event studies**

# A BORDER SEEN FROM SPACE

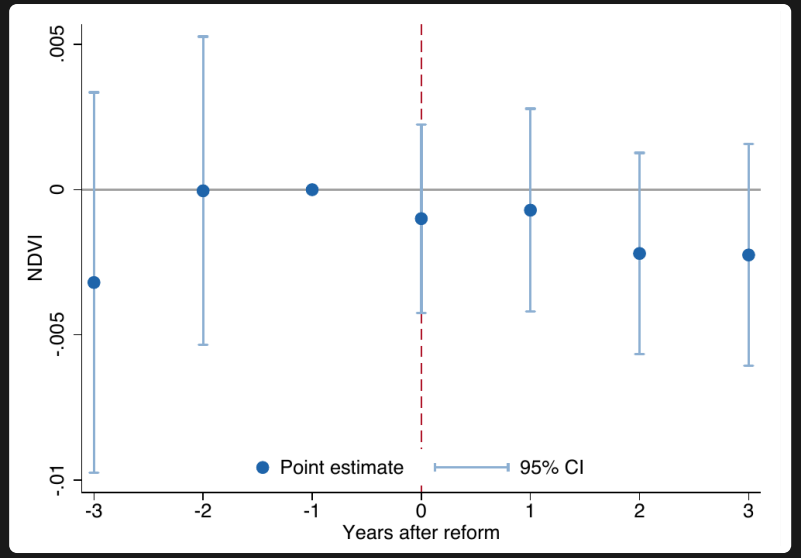



Ferguson and Kim (2026), Fig. 5: vegetation over Anhui and Jiangsu in 1978 and 1982; change in vegetation against distance to the Anhui border

- Anhui reformed first (1978), Jiangsu later. Compare fields **just across the border**, alike in soil and weather, **before and after** the reform: a difference in differences between neighbours
- If the reform raised yields, the Anhui side would have grown greener than the Jiangsu side: the right panel would show a **step at zero**. It is flat

# OFFICIAL DATA AND THE VIEW FROM SPACE

- **Lin (1992)**, official provincial data: the reform explains **almost half** of farm output growth, 1978-84
- **Ferguson and Kim**: vegetation measured by **satellites** (Landsat, from 1972), not by the statistical office
- Same staggered timing, before and after: **no effect on yields**, even three years later
- What moved yields? Their evidence points to the **1979 price increase**; the reform instead freed labour for other work
- Data are political, again: the miracle's first chapter is being **rewritten with new data**




Ferguson and Kim (2026), Fig. 6: change in the border gap in vegetation (NDVI) after a province reforms, years 0-3

# BACK TO THE TRANSITION

- We still cannot say what "the transition" caused. We can say something about its **pieces**:
- Sent-down years: **no lasting benefit** once siblings are compared
- Land reform: **more boys** born after a first girl; on yields, a **live debate** between official data and satellites, with prices as the rival
- Satellites can **overturn** a story built on official data
- Every method buys an answer with an **assumption**. The job is to say which, and to check it

# QUESTIONS, FOR NEXT WEEK

**Two papers, two briefs, one set of questions each**:

- Almond, Li, Zhang (2019): land reform and the missing girls
- Chen, Fan, Gu, Zhou (2020): city youths in village schools

Read the brief (the paper if you like) and answer the questions: causal graphs, the ideal experiment, data, China, theory.

Please no AI chatbots: use the papers, the lectures and, only if needed, a search without AI (google.com/search?udm=14)



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**Send your answers to francesco.manaresi@bancaditalia.it**

**They won't be graded!**